Neil Malan

n.malan@unsw.edu.au

EDUCATION

UNIVERSITY OF CAPE TOWN
BSc (Hons, 1st class), Marine Biology and Ocean and Atmos. Science
Dec 2011 | Cape Town, South Africa

MSc, Ocean and Climate Dynamics
Jul 2013 | Cape Town, South Africa

PhD, Physical Oceanography
Jul 2018 | Cape Town, South Africa

LINKS

Website:// nmalan.github.io
Github:// **nmalan**
LinkedIn:// **neil-malan**

RESEARCH PROFILE

I am an oceanographer with a background in boundary currents, eddies, cross-shelf exchange and mixing – and the observational and modelling methods used to quantify them. Increasingly I am drawn to the ecological side of these systems, where physical structure sets the conditions for productivity, habitat and dispersal. I work especially on marine extremes, including marine heatwaves and flood-driven extreme low salinity events, and on the methods and conceptual framing that link them to biological outcomes. Much of my contribution lies in connecting scales and disciplines – relating basin-scale variability to local coastal impact, and working across the boundaries between oceanography, ecology, fisheries science and coastal hazards.

EXPERIENCE

UNIVERSITY OF NEW SOUTH WALES | Senior Research Associate, Climate Change Research Centre
July 2023 - Present | Sydney, Australia

- Academic level B position, working with Prof. Alex Sen Gupta
- Research focus on subsurface marine heatwaves.
- Research focus on coastal marine heatwaves and their impacts.

UNIVERSITY OF NEW SOUTH WALES | Postdoctoral research fellow, Coastal and Regional Oceanography Laboratory
Nov 2018 - June 2023 | Sydney, Australia

- Academic level A postdoctoral position in the School of Biological, Earth and Environmental Sciences, working with Professor Moninya Roughan.
- Research focus on cross-shelf Western Boundary Current processes and temperature trends, as well as shelf chlorophyll dynamics and coastal salinity extreme events.
- Contribute to management of IMOS observation programme as deputy-leader, New South Wales Moorings.

NANSEN-TUTU CENTRE FOR MARINE ENVIRONMENTAL RESEARCH. | Postdoctoral research fellow
February 2018 – Sep 2018 | Cape Town, South Africa

- Assist in development of base HYCOM model for operational data assimilation.
- Supervise post-graduate research student.
- Organise international workshop on modelling the Agulhas Current System.

SOUTH AFRICAN ENVIRONMENTAL OBSERVATION NETWORK | Doctoral research fellow - professional development program
November 2013 – Sep 2017 | Cape Town, South Africa

- Undertake PhD research.
- Contribute marine expertise to national science meetings.
- Participate in educational programmes for underprivileged schools.

TEACHING

UNIVERSITY OF NEW SOUTH WALES | Course Co-convenor

2024 - present | Sydney, Australia

- MSCI3001/MSCI5004 Physical Oceanography, course size approx. 50 students (with Prof. Alex Sen Gupta) – curriculum design, course management, lectures, tutorials and assessment development.
- 2025 overall student satisfaction 5.63/6, well above School and Faculty averages.

RHODES UNIVERSITY | Lecturer, Dept. of Ichthyology & Fisheries Science

2012 - 2013 | Grahamstown, South Africa

- Taught and coordinated the Fisheries Oceanography module for 2nd- and 3rd-year students, course size approx. 20 students.

TEACHING ASSISTANT | University of Cape Town & Sydney Institute of Marine Science

2014 - 2017; 2022

- Ocean Dynamics (2nd year) and Operational Oceanography (MSc), UCT.
- MSCI6681 Topics in Australian Marine Science (MSc), SIMS.

RESEARCH STUDENTS

COMPLETED

Fernando Sobral. PhD, UNSW 2026
Cross-shelf heat transport in the East Australian Current
Clea Welch. MSc, UCT 2026
Siying Liu - Taught MSc thesis UNSW 2026 - Modelling sea urchin *Centrostephanus rodgersii* larval dispersal from mainland Australia to Tasmania under EAC intensification.
Subsurface marine heatwaves in the Mozambique Channel
Yotam Fadida. MSc, UCT 2019
Trends in the Agulhas Return Current (published in DSR)
Caitlin Sole. Hons, UCT 2016
Impact of relative and absolute winds on eddy variability.

CURRENT

Emma Hu - summer research intern (UNSW) - 1D biogeochemical modelling of the impacts of MHW vertical structures.
Erika Lehmann - Honours (UNSW) - A novel Lagrangian-Eulerian Framework to examine the potential migration of various species polewards down the East Australia Current
Rosa Fenhalls - Taught MSc thesis (UNSW) - A machine learning approach to predicting salinity in the Hawkesbury estuary.
Ronak Rathore - Research Internship (UNSW) - Drivers and impacts of the recent Java Marine Heatwave.
Leon Jin - Honours (UNSW) - The hidden role of vertical mixing in driving Marine Heatwaves.

AWARDS

2023 Invited Speaker, *Gordon Research Seminar on Coastal Ocean Dynamics*
2021 Best Postdoctoral Research Paper, *UNSW CMSI*
2018 Invited participant, Physical Oceanography Dissertation Symposium, *National Science Foundation*
2016 Best Student Presentation, Agulhas Current Research Symposium, *SAEON*
2015 SPACES travel fellowship (2-month funded research visit), *GEOMAR, Kiel*

ACADEMIC SERVICE AND OUTREACH

- **Member** 2023-present, Marine Heatwaves International Working Group .
- **Member of Australia's Marine National Facility scientific advisory committee** 2024-present, provide expert advice to the Marine National Facility (MNF) Steering Committee, primarily focused on assessing the research quality of proposed MNF voyages and other matters requiring scientific expertise.
- **Convenor** Oceania Marine Heatwave Seminars. Jan 2025 - present
- **Member of the scientific committee**, 2026 International Conference on Southern Hemisphere Meteorology and Oceanography.
- **Invited contributor**, three chapters (one as co-lead), AGU monograph on marine heatwaves and their impacts (in preparation).
- **Deputy Leader IMOS NSW moorings 2018-2023**, assist in management of mooring program, including contribution to national committees, meeting reporting requirements at both state and national levels, tracking impact and developing data analysis tutorials.

- **Session Convenor or co-convenor** for sessions on Australian Boundary Currents at the Australian Meteorological and Oceanographic Society Annual meetings in 2020 and 2022, as well as Subtropical Western Boundary Current - Shelf interactions at the Ocean Sciences Meeting in 2022.
- **Expert Reviewer** for over 50 articles in journals such as *Geophysical Research Letters*, *Nature Climate Change*, *Journal of Physical Oceanography*, *Global Biogeochemical Cycles* and *Geoscientific Model Development*.

PUBLICATIONS

I have published widely in top-tier oceanography and climate journals, with 10 first-author and 5 second-author papers. My work has been cited 657 times with an h-index of 14 (Google Scholar, 29/10/2026).

[1] Neil Malan, Moninya Roughan, Michael Hemming, Neil Doszpot, Colin L Johnson, Samantha K Lynch, Robert A B Mason, and Tim Ingleton. "Observations of river plumes in southeastern Australia impacting the coastal ocean following extreme rainfall in 2022". In: *Marine and Freshwater Research (in press)* (2026).

[2] Neil Malan, Alex Sen Gupta, Amandine Schaeffer, Shujing Zhang, Martina A Doblin, Gabriela Semolini Pilo, Andrew E Kiss, Jason D Everett, Erik Behrens, Antonietta Capotondi, et al. "Lifting the lid on Marine Heatwaves". In: *Progress in Oceanography* (2025), p. 103539.

[3] Kathryn E Smith, Alex Sen Gupta, Dillon Amaya, Jessica A Benthuysen, Michael T Burrows, Antonietta Capotondi, Karen Filbee-Dexter, Thomas L Frölicher, Alistair J Hobday, Neil J Holbrook, et al. "Baseline matters: Challenges and implications of different marine heatwave baselines". In: *Progress in Oceanography* 231 (2025), p. 103404.

[4] Kathryn E. Smith, Alex Sen Gupta, Michael T. Burrows, Karen Filbee-Dexter, Alistair J. Hobday, Neil J. Holbrook, Neil Malan, Pippa J. Moore, Eric C. J. Oliver, Mads Thomsen, Thomas Wernberg, Zijie Zhao, and Dan A. Smale. "Ocean extremes as a stress test for marine ecosystems and society". In: *Nature Climate Change* 15.3 (2025), pp. 231–235.

[5] Clea B Welch, Neil Malan, Daneeja Mawren, Tamaryn Morris, Janet Sprintall, and Juliet C Hermes. "Subsurface manifestation of marine heat waves in the southwestern Indian Ocean". In: *Ocean Science* 21.4 (2025), pp. 1695–1708.

[6] Neil Malan, Moninya Roughan, Michael Hemming, and Tim Ingleton. "Quantifying coastal freshwater extremes during unprecedented rainfall using long timeseries multi-platform salinity observations". In: *Nature Communications* 15.1 (2024), p. 424.

[7] Fernando Sobral, Moninya Roughan, Neil Malan, and Junde Li. "Mean-state and seasonal variability in temperature structure and heat transport in the East Australian Current System from a multi-decadal regional ocean model". In: *Journal of Geophysical Research: Oceans* 129.9 (2024), e2023JC020438.

[8] Michael P Hemming, Moninya Roughan, Neil Malan, and Amandine Schaeffer. "Observed multi-decadal trends in subsurface temperature adjacent to the East Australian Current". In: *Ocean Science* 19.4 (2023), pp. 1145–1162.

[9] Neil Malan, Moninya Roughan, Michael Hemming, and Amandine Schaeffer. "Mesoscale circulation controls chlorophyll concentrations in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 128.4 (2023), e2022JC019361.

[10] Neil Malan, Moninya Roughan, Geoffrey J Stanley, Ryan Holmes, and Junde Li. "Quantifying cross-shelf transport in the East Australian Current system: a budget-based approach". In: *Journal of Physical Oceanography* 52.10 (2022), pp. 2555–2572.

[11] Lachlan R Phillips, Neil Malan, Moninya Roughan, Robert Harcourt, Ian Jonsen, Martin Cox, Andrew S Brierley, David Slip, Adam Wilkins, and Gemma Carroll. "Coastal seascape variability in the intensifying East Australian Current Southern Extension". In: *Frontiers in Marine Science* (2022).

[12] Hayden T Schilling, Daniel E Hewitt, Neil Malan, Matthew D Taylor, and Daniel D Johnson. "Cross-jurisdictional larval supply essential for eastern Australian spanner crabs (Ranina ranina)". In: *Marine and Freshwater Research* 73.11 (2022), pp. 1352–1367.

[13] Yotam Fadida, Neil Malan, Meghan F Cronin, and Juliet Hermes. "Trends in the Agulhas Return Current". In: *Deep Sea Research Part I: Oceanographic Research Papers* 175 (2021), p. 103573.

[14] Neil Malan, Moninya Roughan, and Colette Kerry. "The rate of coastal temperature rise adjacent to a warming western boundary current is nonuniform with latitude". In: *Geophysical Research Letters* 48.3 (2021), e2020GL090751.

[15] Neil Malan, Matthew Archer, Moninya Roughan, Paulina Cetina-Heredia, Michael Hemming, Carlos Rocha, Amandine Schaeffer, Iain Suthers, and Eduardo Queiroz. "Eddy-driven cross-shelf transport in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 125.2 (2020), e2019JC015613.

[16] Jethan Shaun d'Hotman, Neil Malan, Charine Collins, Marc de Vos, Rick Lumpkin, Tamaryn Morris, and Juliet Hermes. "The use of a jet reference frame to analyze drifter trajectories in the Agulhas Current". In: *Journal of Geophysical Research: Oceans* 124.6 (2019), pp. 4238–4247.

[17] Neil Malan, Jonathan V Durgadoo, Arne Biastoch, Chris Reason, and Juliet Hermes. "Multidecadal wind variability drives temperature shifts on the Agulhas Bank". In: *Journal of Geophysical Research: Oceans* 124.5 (2019), pp. 3021–3035.

[18] Ahmad Fehmi Dilmahamod, B Aguiar-González, Pierrick Penven, CJC Reason, WPM De Ruijter, N Malan, and JC Hermes. "SIDDIES corridor: a major east-west pathway of long-lived surface and subsurface eddies crossing the subtropical South Indian Ocean". In: *Journal of Geophysical Research: Oceans* 123.8 (2018), pp. 5406–5425.

[19] N. Malan. "Are icebergs a realistic option for augmenting Cape Town's water supply?" In: *Water Wheel* 17.2 (2018), pp. 32–34.

[20] Neil Malan, Björn Backeberg, Arne Biastoch, Jonathan V Durgadoo, Annette Samuelsen, Chris Reason, and Juliet Hermes. "Agulhas Current Meanders facilitate shelf-slope exchange on the Eastern Agulhas Bank". In: *Journal of Geophysical Research: Oceans* (2018).

[21] N Malan, CJC Reason, and BR Loveday. "Variability in tropical cyclone heat potential over the Southwest Indian Ocean". In: *Journal of Geophysical Research: Oceans* 118.12 (2013), pp. 6734–6746.

PREPRINTS, IN-REVIEW AND SUBMITTED PAPERS

- Karen Filbee-Dexter, Lianna Gendall, Albert Pessarrodona, Michael T Burrows, Alex Sen Gupta, Alistair J Hobday, Neil J Holbrook, Neil Malan, Shinae Montie, Kira A Krumhansl, Pippa J Moore, Erik C J Oliver, Dan A Smale, Kathryn E Smith, Samuel A Starko, Mads S Thomsen, Mathew A Vanderklift, Zijie Zhao, and Thomas Wernberg. "Marine heatwaves drive widespread loss and slow recovery of coastal carbon cycling". In: *Communications Earth & Environment (submitted)* (2026). Preprint posted on Research Square, 28 May 2026. doi: `10.21203/rs.3.rs-9834051/v1`.

- Ryan M Holmes and Neil Malan. "Accurately diagnosing mixed layer tracer budgets and marine heatwave drivers in ocean models". In: *Journal of Advances in Modeling Earth Systems (submitted)* (2026). Preprint on ESSOAr; submitted 10 June 2026.

- Moninya Roughan, Robert A. B. Mason, Marina Azaneu, David Gwyther, Michael Hemming, Colette Kerry, Junde Li, Neil Malan, Amandine Schaeffer, Fernando Sobral, Natacha Bourg, and Connor Henderson. "The imprint of global change in the East Australian Current System: non-uniform trends and extremes in an ocean warming hotspot". In: *Reviews of Geophysics (in revision)* (2026).

- Kathryn E Smith, Julio Salcedo-Castro, Neil J Holbrook, Serge Andrefouet, Hans Burchard, Michael T Burrows, Bronwyn Cahill, Sophie Cravatte, Karen Filbee-Dexter, David Carrasco, Christopher D G Harley, Alistair J Hobday, Nathan G King, Romain Le Gendre, Neil Malan, Murilo Zanetti Marochi, Cristian Martinez-Villalobos, Piero L F Mazzini, Pippa J Moore, Antonio Olita, Eric Oliver, Andrea Piñones, Cécile Pujol, Alex Sen Gupta, Spencer Tassone, Mads S Thomsen, Ryan K Walter, Thomas Wernberg, and Dan A Smale. "Marine heatwaves in coastal systems". In: *Nature Reviews Earth & Environment (in review)* (2026).

- Zijie Zhao, Karen Filbee-Dexter, Eric C. J. Oliver, Michael T. Burrows, Alistair J. Hobday, Neil J. Holbrook, Neil Malan, Shinae Montie, Pippa J. Moore, Alex Sen Gupta, Dan A. Smale, Mads S. Thomsen, Thomas Wernberg, and Kathryn E. Smith. "Accelerating marine heatwaves in the 21st century". In: *Nature Communications (in revision)* (2026).

- Zijie Zhao, Alistair J. Hobday, Neil J. Holbrook, Alex Sen Gupta, Eric C. J. Oliver, Lukas Brunner, Michael T. Burrows, Karen Filbee-Dexter, Neil Malan, Shinae Montie, Pippa J. Moore, Dan A. Smale, Kathryn E. Smith, Mads S. Thomsen, and Thomas Wernberg. "Detecting temperature extremes in a statistically optimal way". In: *Journal of Climate (submitted)* (2026).